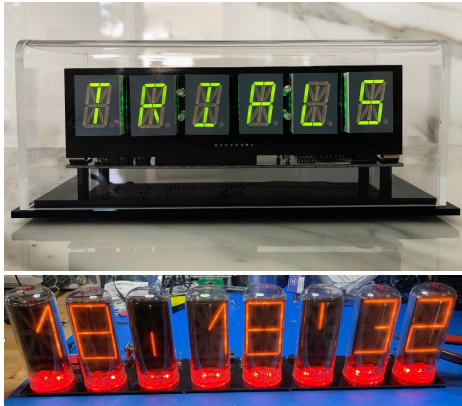


WordClock 1–4

Word Clock System

Assembly Manual



April 2026

This manual describes the assembly, testing, and bring-up of the WordClock 1, 2, 3, and 4 family from bare PCB and components.

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Designed by Mitchell Feig — Angry Electronics

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1. Introduction

This manual covers the assembly of the WordClock 1, 2, 3, and 4 family of alphanumeric word clocks. All four hardware variants run the same firmware (described in the Operator Manual) but use different display technologies and PCB layouts:

- WordClock 1 — 0.8" 16-segment alphanumeric LED, 6/8/16 digits
- WordClock 2 — 2.3" 16-segment alphanumeric LED, 6/8/16 digits
- WordClock 3 — IV-17 vacuum-fluorescent display tubes, 8 or 16 digits
- WordClock 4 — Burroughs B7971 Nixie tubes, 6 or 8 digits

Each clock's assembly procedure is in its own chapter. Read the chapter that matches the kit being built. The Introduction, Assembly Notes, Tools and Equipment, and Safety chapters apply to every variant.

These kits are intended for experienced electronics builders. Although construction is relatively straightforward, every kit includes surface-mount components that demand a steady hand, good lighting, and patience. If in doubt, contact the distributor before opening the kit — kits may be returned for a full refund (less shipping) within 30 days of purchase if assembly has not been started.

2. Assembly Notes

Read this entire chapter before beginning assembly of any variant.

Case and PCB Planning

If a custom case will be built for the clock, use the bare PCB as a template for hole locations and overall dimensions before installing components. Once components are soldered the board is no longer a clean reference.

If the PCBs are supplied as a single panel with a V-score between display and main boards, separate them by snapping the panel apart. Lay the larger of the two against a flat surface like a workbench and apply downward pressure on the smaller side. The V-score should snap cleanly. Snap before installing components.

Component Verification

Measure every resistor with a multimeter before installing it. Do not trust eyesight alone to read color codes — tracing a misplaced resistor after the rest of the board is populated is time-consuming and easy to avoid.

Polarized parts (electrolytic capacitors, diodes, integrated circuits, LEDs) are clearly marked on the silkscreen. Observe orientation. If a marking is unclear, contact the distributor before soldering.

Surface-Mount Soldering

A flux pen is essential for SMD components. The standard procedure is:

- Apply a small amount of solder to one corner pad on the footprint.
- Place the component, then re-flow the corner pad to tack it in place.
- Verify alignment of every pin against its pad. Adjust if needed.
- Solder the diagonally opposite pin to lock the part in place.
- Verify alignment a second time, then solder the remaining pins.
- Inspect under magnification for solder bridges and cold joints. Use solder wick to clean any bridges.

Power-Up Discipline

Each chapter's "Step 1" instructs builders to install only the power-supply components first, then verify the test-point voltages with a multimeter before installing the rest of the parts. Do not skip this step. A short or wrong-voltage rail can destroy ICs in seconds. If a test point reads outside the expected range, stop and find the cause before continuing.

Power Supply

All four variants run from a 12V DC center-positive supply with a 5.5 × 2.1 mm barrel plug. The smaller boards (WordClock 1, WordClock 4 6-tube) need at least 1A; the larger LED, VFD, and 8-tube Nixie boards need 2A or more. Use a quality regulated UL-listed adapter — there is little point in carefully assembling a precision clock and then powering it from a noisy throwaway supply.

Schematics

Each variant's schematic sheets are documented in a separate WC1-4 Schematics PDF (available from the distributor and the StocksClocks website). Keep the schematics open as a reference while assembling — they show every signal name, IC pinout, and component value at a readable size.

3. Required Tools and Equipment

Soldering

- Temperature-controlled soldering iron (recommended 350°C / 660°F)
- Fine solder (0.5–0.8 mm rosin-core, 60/40 lead or lead-free)
- Liquid or paste solder flux, plus a flux pen for SMD work
- Solder wick and/or a desoldering pump
- Fine-tip tweezers for SMD components
- Magnifying visor or bench magnifier (essential for the HT16K33 and HV5530 packages)

Test Equipment

- Digital multimeter (must measure DC voltage up to 200 V for WordClock 4)
- 12 V DC power supply, 1 A minimum for testing (2 A for WordClock 2, 3, and the 8-tube WordClock 4)
- Oscilloscope (optional but useful for tracing data and clock signals)

Hand Tools

- Small Phillips and flathead screwdrivers
- Wire strippers and flush cutters
- Diagonal cutters (for trimming Nixie / VFD tube leads)
- Needle-nose pliers (for removing pins from black plastic header strips)
- Anti-static wrist strap
- Masking tape (for holding components in alignment during soldering)

4. Safety Warnings

WARNING: WordClock 4 (Nixie) generates approximately 170 V DC for the Nixie tube anodes. This voltage is lethal. Never touch the high-voltage circuit while the clock is powered on. Always disconnect power and wait at least 30 seconds for storage capacitors to discharge before working on the board.

WARNING: WordClock 3 (VFD) generates roughly 35 V on the tube anode rail and 2.4 V on the filament rail. The 35 V rail is not lethal but can damage components and produce a noticeable shock. Use the same disconnect-and-discharge discipline.

WARNING: Nixie and VFD tubes are glass and fragile. Handle the tubes by their bases only. Do not apply excessive force when seating tube leads into PCB sockets.

CAUTION: Observe correct polarity on every electrolytic capacitor (C1, C2, C9, etc.), diode, and IC. Reversed polarity destroys components instantly.

CAUTION: Linear and switching regulator ICs (LM2576) become hot during operation. Allow adequate ventilation; do not touch the metal tab during operation.

CAUTION: Some kits supply two LM2576 variants — one fixed (LM2576T-5.0) and one adjustable (LM2576-ADJ). They are NOT interchangeable. Read the silkscreen carefully before installing each.

5. WordClock 1 — Small LED



Finished WordClock 1 (6-digit, green LED).

WordClock 1 uses 0.8" alphanumeric LED displays driven by an HT16K33 LED driver IC. The kit ships in 6-digit, 8-digit, and 16-digit configurations; the parts list below covers the 6-digit version. Refer to the schematics PDF for the 8- and 16-digit display board variations (additional HT16K33 driver, additional LED rows).

5.1 Parts List

Reference	Qty	Part	Value	Notes
R1, R2	2	Resistor	4.7k $\frac{1}{4}$ W	
R3, R4	2	Resistor	10k $\frac{1}{4}$ W	
R5	1	Resistor	39k $\frac{1}{4}$ W	
R6–R9	4	Resistor SMD	470 Ω or 315 Ω	470 Ω for Orange / Super Red / Light Green; 315 Ω for others
C1, C2	2	Capacitor	330 μ F 25 V	Electrolytic — note polarity
C3–C8	6	Capacitor	100 nF	
C9	1	Capacitor	100 nF	SMD 1210, marked 104
D1	1	Diode	1N4001	Reverse-polarity protection
D2, D3	2	Diode	1N5819	Schottky
F1	1	Poly Fuse	500 mA	Labeled .050 (supplied part may vary)
J1	1	Power Jack	2.1 mm	
J2	1	SD Card Jack	DM1AA-SF-PEJ	

J3	1	Audio Jack	3.5 mm	SJ-3523-SMT-TR (GPS)
J4	1	Header	1×8 2.54 mm	Female
J5	1	Header	1×8 2.54 mm	Male, 90°
J6, J7	2	Header	1×4	Not used
—	1	Header strip	1×40	Cut into two 19-pin rows for U2
L1	1	Inductor	100 µH	
LED1–LED6	6	LED	0.8" 16-seg	Alphanumeric display
LED7–LED10	4	LED	3 mm	Colon LEDs, color matched
LED11–LED16	6	LED	APA102 SMD	RGB backlight
Q1	1	Sensor	TEPT4400	Ambient light (looks like a clear LED)
PZ1	1	Buzzer	11 mm pitch	Piezo
SW1, SW2	2	Pushbutton	90°	
U1	1	Regulator	LM2576T-5.0	5 V switching
U2	1	Microcontroller	ESP32 PICO-KIT	Pre-programmed with WordClock firmware
U3	1	IC	74AHC125D	Quad buffer, SMD
U4	1	IC	HT16K33	LED driver, SOP-28
—	108	Pin	Round female	For 16-segment displays
—	2	Bracket	Keystone 4326	
—	4	Standoff	10 mm	Threaded
—	6	Screw	M3 8 mm	
—	2	Nut	M3	
—	1	Silver tape	—	Cut to cover the back of each display (optional)
—	1	SD card	—	Pre-loaded with WordClock library files
—	1	PCB	—	Display + Main panel, V-scored

5.2 Step 1 — Build the Main Board

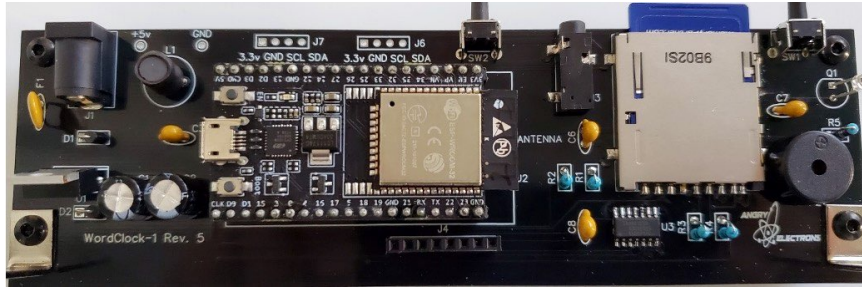
Build the power supply first. Install J1, F1, C1, C2, C3, D1, D2, L1, and U1. C1, C2, D1, and D2 must be installed in the orientation marked on the silkscreen.

Connect a 9–12 V DC supply with a 2.1 mm center-positive plug to J1 and check the test point for 5 V. Do not continue assembly if 5 V is not present.

Install the remaining components on the main board. Begin with U3 (74AHC125D, SOIC-14). The dot or notch on the chip must align with the cutout on the silkscreen footprint. Use the SMD soldering procedure from Section 2.

Install the SD card connector next. The three small leads on the side closest to SW1 do not need to be soldered. Plastic chassis on these connectors is easily damaged by excess heat — work quickly. Install J3 (GPS connector).

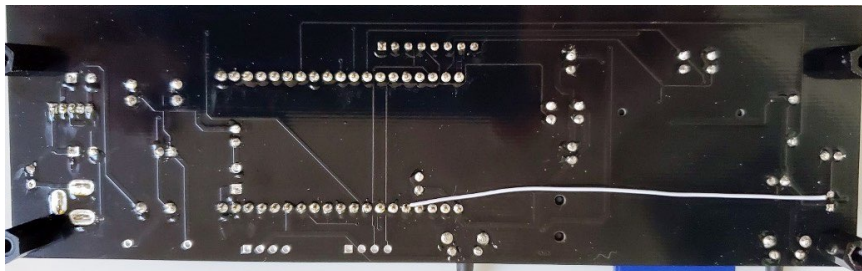
The ambient light sensor Q1 should sit about 10 mm above the board. The shorter lead goes toward the back of the board.



WordClock 1 main board (top), showing the ESP32 socket area and SD card connector.

Cut the 40-pin female header strip into two 19-pin pieces and install at the U2 location. Do NOT solder the microcontroller to the board — it plugs in afterwards.

Install the remaining parts on the main board. J6 and J7 are not used.

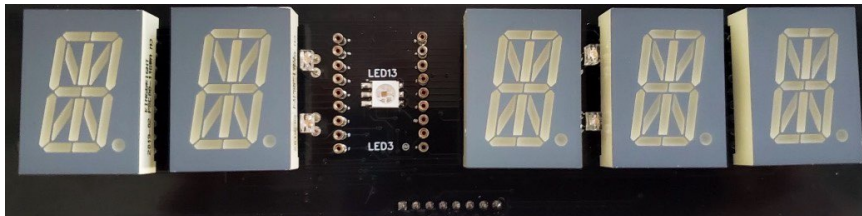


WordClock 1 main board (bottom), showing the wire jumper that connects Q1 to ESP32 IO34.

On the bottom of the board, run a jumper wire from the back of Q1 (the TEPT4400 pin farthest from the switch immediately behind it) to pin 5 of the U2 microcontroller socket — labeled IO34 or G34 on the silkscreen.

Insert the microcontroller (U2) into the headers, aligning the USB connector and the antenna with the marks on the silkscreen.

5.3 Step 2 — Build the Display Board



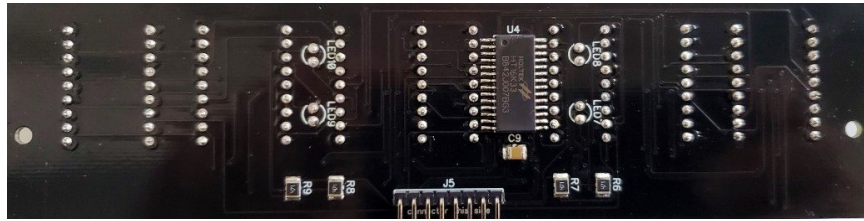
WordClock 1 display board (front), with backlight LEDs and colon LEDs installed.

Install LED11–LED16 (APA102 SMD RGB backlights). Use the standard SMD procedure. Note orientation — align the angled corner with the mark on the board.

Install the four 470 Ω SMD resistors at R6–R9. These equalize brightness between the displays and the colons; if a different colon color is used, this value may need adjustment.

Install C9 (the SMD ceramic capacitor) on the back of the display board.

Install J5 (90° male header) on the back of the display board. Solder one pin first, then verify the pins are parallel to the board before completing.



WordClock 1 display board (back), with U4 (HT16K33) and the male header J5 installed.

Cut the silver tape into six pieces and stick one piece behind each alphanumeric display (LED1–LED6). Verify the tape does not touch any pins. The displays are translucent — without the tape, backlight bleeds through unlit segments. The tape's reflectivity also boosts overall backlight visibility.

Do NOT install the round pins still pressed into their black plastic carrier strips. Pop them out individually and install one round pin per display pin. Hold each black header upside down on the sharp corner of a workbench, grip the back of a pin with needle-nose pliers, and push down hard — the pin will pop free.

Install one round pin into each pin position on LED1–LED6. Use masking tape to hold the displays in place. Position the display board upside down on a flat surface and solder all pins.

Solder U4 (HT16K33) to the back of the display board, observing orientation. Use the standard SMD procedure.

Install LED7–LED10 (the colon LEDs). The short pin goes toward the top of the display board. Align the LEDs so their tops sit even with the front face of the alphanumeric displays.

5.4 Step 3 — Final Assembly

Install two 10 mm standoffs into the holes on the bottom of the main board on the side with the buttons (use two screws).

Install two more standoffs into the holes on the front edge, with the short side of the 90° brackets sandwiched between bracket and board: screw → bracket short side → main board → standoff.

Plug the display board into the main board, then secure with two screws and nuts through the brackets. The display board sits in front of the brackets.

Power on, configure WiFi via the AP setup form, and proceed to the Operator Manual for setup and operation.

6. WordClock 2 — Large LED



Finished WordClock 2 in the 16-digit configuration.

WordClock 2 uses 2.3" alphanumeric LED displays — the same family as WordClock 1 but four times larger. The larger displays demand more drive current, so the board carries an additional 10.5 V switching rail (LM2576-ADJ) and current-balancing transistor arrays (TBD62783, ULN2803). The kit ships in 6-, 8-, and 16-digit configurations; the parts list below covers all three with quantity columns.

6.1 Parts List (6-, 8-, and 16-digit)

Reference	Qty	Part	Value	Notes
C1, C2, C9	3	Capacitor	330 μ F 25–50 V	Electrolytic — note polarity
C3–C8, C10	7	Capacitor	100 nF	Through-hole
C11–C14	4	Capacitor	100 nF	SMD
D1	1	Diode	1N4001–1N4007	Supplied value may vary
D2–D4	3	Diode	1N5819	Schottky — note polarity
F1	1	Poly Fuse	0.5–1 A	
J1	1	Power Jack	2.1 mm	
J2	1	SD Card Jack	DM1AA-SF-PEJ	
J3	1	GPS Jack	SJ-3523-SMT-TR	3.5 mm SMD
J4	1	Header	HDR-1 $\times$ 10	Female
J5	1	Header	HDR-1 $\times$ 10	Male
L1, L2	2	Inductor	100 μ H	

LED1–LED6 / LED1–LED8	6 or 8 (or 2×8)	Display	2.3" 16-seg CC	
LED7–LED10	4	Colon LED	5 mm	Six-digit version only
LED11–LED34 / LED11–LED42	24 / 32	Backlight	APA102	Six- or eight-digit
PZ1	1	Buzzer	—	
Q1	1	Sensor	TEPT4400	Light sensor
R1, R2 (processor)	2	Resistor	4.7k	
R1, R2 (display)	2	Resistor	330Ω	
R3, R4	2	Resistor	10k	
R5	1	Resistor	39k	
R6–R9	4	Resistor SMD	230Ω or 315Ω	Color-matched to display
R10–R12	3	Resistor array	4.7k	
R13, R14, R17, R18, R24, R28	6	Resistor SMD	270Ω or 180Ω	Color-matched
R15, R16, R19–R23, R25–R27	10	Resistor SMD	150Ω or 90.9Ω	Color-matched
R29	1	Resistor	9k	
R30	1	Resistor	1.2k	
SW1, SW2	2	Pushbutton	90°	
U1	1	Regulator	LM2576T-5.0	NOT interchangeable with U8
U2	1	Microcontroller	ESP32 PICO-KIT	Pre-programmed
U3	1	IC	74HC14	
U4	1	IC	HT16K33	
U5, U6	2	IC	TBD62783	
U7	1	IC	ULN2803	
U8	1	Regulator	LM2576-ADJ	NOT interchangeable with U1
U9, U10	2	IC	74HC04D	8-digit display only
—	1	Header strip	1×40	Cut to 2×17 for U2
—	2	Bracket	Keystone 4326	
—	8	Standoff	10 mm	
—	10	Screw	M3 8 mm	
—	2	Nut	M3	
—	1	SD card	—	Pre-loaded with libraries

6.2 Step 1 — Build the Main Board



WordClock 2 main board (top), showing both LM2576 regulators (U1 and U8), the SD card connector, and the ESP32 socket.

Build the 5 V supply first. Install J1, F1, C1, C2, C3, D1, D2, L1, and U1. C1, C2, D1, and D2 must be oriented per the silkscreen.

Connect a 12 V DC center-positive supply to J1 and check the test point for 5 V. Do not proceed if 5 V is not present.

Build the 10.5 V supply. Install C9, C10, D4, R29, R30, L2, and U8. C9 and D4 must be oriented per the silkscreen.

Reconnect the supply and verify the test point reads approximately 10.5 V. Do not proceed if it does not.

CAUTION: Note that the LM2576 marked "ADJ" must be installed at U8 — the fixed -5.0 part installs at U1. The two regulators look similar; misplacement will burn out the display.

Install the remaining components on the main board. Save Q1 (light sensor) for last. Begin with U3 (74AHC125D, SOIC-14) and follow with the other SMD parts.

Install the SD card connector. The three small leads on the SW1 side need not be soldered. Plastic parts on the connector are heat-sensitive — work quickly. Install J3 (GPS connector).

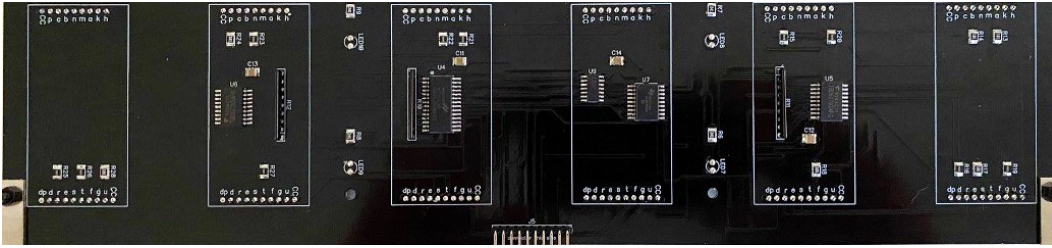
Install Q1 about 10 mm above the board, with the shorter lead toward the back.

Cut the 40-pin female header into two 17-pin pieces and install at the U2 location. Do NOT solder the microcontroller into place.

Install the remaining parts. J6 and J7 are not used.

Insert U2 into the headers, aligning the USB connector and the antenna with the silkscreen marks.

6.3 Step 2 — Build the Display Board



WordClock 2 display board (back), showing the HT16K33 driver and TBD62783 / ULN2803 current-driver ICs.

All parts go on the back (silkscreen side) of the display board, except the backlight LEDs and the alphanumeric displays themselves.

Install LED11–LED34 (APA102 backlights). Align the angled corner with the upper-left silkscreen mark. If an LED needs to be removed, cut it horizontally with diagonal cutters to expose the pins, then unsolder each pin individually.

Install SMD resistors R6–R9. The exact value depends on the display color; the supplied resistors are pre-matched. These equalize brightness between the displays and colons.

Install C11–C14 (SMD ceramic capacitors, usually unmarked tan parts).

Install J5 (90° male header) on the back of the display board. Solder one pin first, verify the header is parallel to the board, then solder the remaining pins.

Cut the pin headers to size for LED1–LED6. Each display uses a 9-pin header on top and a 10-pin header on the bottom. Tape the headers in place, position the board upside down on a flat surface, and solder one pin per header. Verify the headers are perpendicular to the board, then solder the remaining pins.

If individual round pins are supplied instead of strips, install them on each display before soldering. Tape the displays in place, then solder all pins.

Solder U4–U7 to the back of the board. Note the orientation — pin 1 is up on all ICs except U6.

Solder resistor arrays R10–R12 to the back. The mark on each array must align with the square side of the silkscreen footprint. Pin 1 is up on R10 and R11, down on R12.

Six-digit version only — install LED7–LED10 (colon LEDs). Do not solder yet. Short pin goes toward the top of the display board. Flip the board face-down on the workbench, ensure the LEDs touch the work surface, and solder one lead per LED. Verify alignment, then solder the remaining lead per LED.

6.4 Step 3 — Final Assembly

Install four 10 mm standoffs into the back-edge holes on the main board (button side), using four screws.

Install two more standoffs into the front-edge holes (outer corners) with the short side of each 90° bracket sandwiched between bracket and board. Install screws and standoffs into the remaining six holes.

Plug the display board into the main board with the brackets behind the display board. Secure with two screws and nuts through the brackets.

Remove the protective film from each alphanumeric display.

Power on and proceed to the Operator Manual for setup.

7. WordClock 3 — VFD



Finished WordClock 3 (8-digit, IV-17 vacuum-fluorescent tubes).

WordClock 3 uses Russian-surplus IV-17 (or IV-4 for 16-digit variants) vacuum-fluorescent display tubes mounted on individual carrier boards. The main board includes the standard 5 V regulator plus a separate ~35 V anode rail and a 2.4 V filament rail. Each tube board carries one VFD and one APA106 backlight LED, and plugs into a 2×6 socket on the main board.

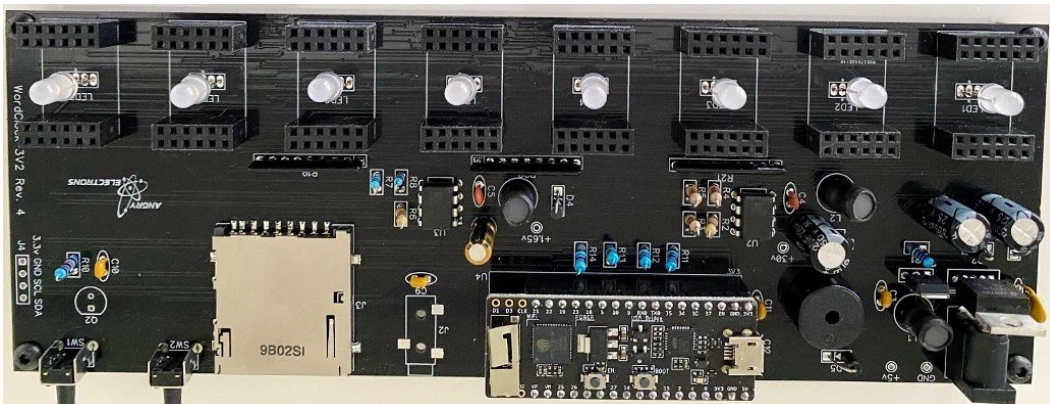
7.1 Parts List — 8-Digit Version

Reference	Qty	Part	Value	Notes
C1, C2	2	Capacitor	330 μ F 25–50 V	Electrolytic
C3, C9–C12	5	Capacitor	100 nF	Through-hole
C4	1	Capacitor	330 pF	
C5	1	Capacitor	180 pF	
C6	1	Capacitor	220 μ F 50 V	
C7	1	Capacitor	100 μ F 16 V	
C8, C13–C18	7	Capacitor	100 nF	SMD
D1	1	Diode	1N4007	
D2–D5	4	Diode	1N5819	Schottky
F1	1	Poly Fuse	0.5–1 A	
J1	1	Power Jack	2.1 mm	
J2	1	GPS Jack	SJ-3523-SMT-TR	3.5 mm SMD
J3	1	SD Card Jack	DM1AA-SF-PEJ	
L1–L3	3	Inductor	100 μ H	
LED1–LED8	8	RGB LED	APA106	One per tube board
PZ1	1	Buzzer	—	
Q1	1	MOSFET	IRFU5305	
Q2	1	Sensor	TEPT4400	Light sensor
R1, R11, R12	3	Resistor	10k	

R2, R6	2	Resistor	0.47Ω	
R3	1	Resistor	180Ω	
R4	1	Resistor	2.7k	
R5	1	Resistor	62k	
R7	1	Resistor	5.1k	
R8	1	Resistor	4.7k	
R10	1	Resistor	120k	
R13, R14	2	Resistor	4.7k	
R19–R21	3	Resistor array	4.7k	
SW1, SW2	2	Pushbutton	90°	
U1	1	Regulator	LM2576T-5.0	5 V switching
U2, U3	2	IC	MC34063AP1	HV switching
U4	1	Microcontroller	ESP32 PICO-KIT	Pre-programmed
U5	1	IC	74AHC125D	
U6	1	IC	HT16K33	
U7, U8, U10	3	IC	TBD62783	
U9, U11	2	IC	74HC04D	
—	4	Standoff	10 mm	
—	4	Screw	M3 8 mm	
—	8	Tube board	IV-17 carrier	
—	8	Header	2×6 male	
—	8	Header	2×6 female	
—	1	Header	1×40 female	For ESP32
—	1	SD card	—	Pre-loaded

The 16-digit version uses substantially the same parts list with quantities scaled, plus an additional HT16K33 (U12), three additional TBD62783 (U13–U16), three additional 74HC04D (U15), and 16 tube boards. See the schematics PDF for the full BOM.

7.2 Step 1 — Build the Main Board



WordClock 3 main board (top), showing the 2×6 female sockets that receive the tube boards.

Install the 2×6 pin female headers on the top of the board.

Cut the 40-pin female header into two 17-pin pieces and install at the U4 location. Do NOT solder the microcontroller to the board.

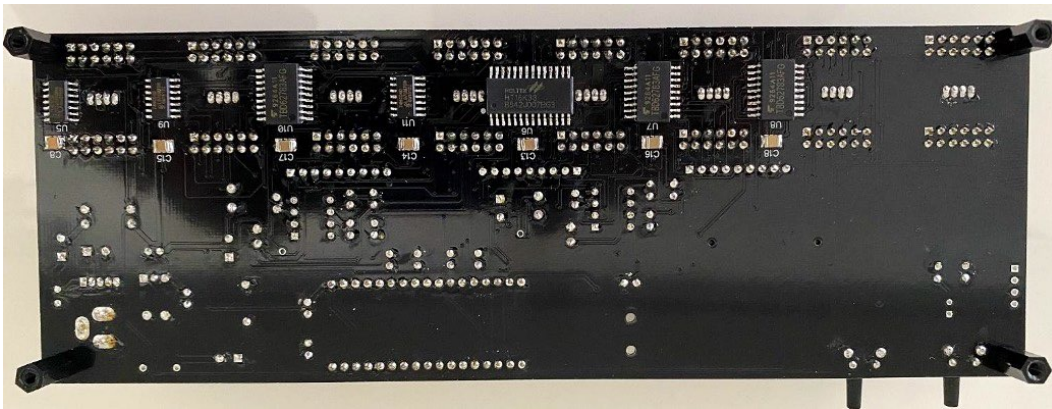
Build the 5 V supply. Install J1, F1, C1, C2, C3, D1, D2, L1, and U1. C1, C2, D1, and D2 must be oriented per the silkscreen.

Connect a 12–18 V supply to J1 and check the test point for 5 V. Do not continue if 5 V is not present.

Build the 35 V and 2.4 V supplies (note the silkscreen voltage markings are outdated). Install R1–R8, L2, L3, D3, D4, C4–C7, Q1, U2, and U3. D3, D4, C6, C7, U2, and U3 must be oriented per the silkscreen.

WARNING: Temporarily connect the ENABLE pad to GND. Do not install the microcontroller while the jumper is in place — applying the high-voltage rail with the ESP32 in place will damage it.

Connect the supply and verify the test points read approximately 35 V and 2.4 V. Do not continue if they do not. Remove the temporary jumper.



WordClock 3 main board (bottom), showing the SMD ICs and SD card connector.

Install the components on the bottom of the board, starting with the surface-mount ICs U5–U11. Use the standard SMD procedure. The 16-digit version has additional ICs to install at this point — refer to the schematics PDF.

Install the remaining SMD parts on the bottom: C8 and C13–C18.

Install the SD card connector. All leads must be soldered to the board. Several pins are very close to the grounded case — be sure they only contact the PCB pads, not the case. Install J3 (GPS connector).

Install all remaining top-side parts except the RGB LEDs (LED1–LED8 / LED1–LED16). The LEDs are installed last.

Install Q2 (light sensor) about 10 mm above the board, shorter lead toward the back.

Install the remaining parts. J4 is not used.

Insert the microcontroller into the headers, aligning the USB connector and antenna with the silkscreen marks.

7.3 Step 2 — Build the Tube Boards

Cut the leads of all IV-17 tubes progressively shorter, making it easier to thread the leads through the carrier-board holes.

Insert the tube leads into the boards, observing the "Front" and "Tube Other Side" silkscreen markings.

Solder one lead of each tube first, then check both front-to-back and side-to-side alignment. Solder a second lead 90° from the first and re-check. If aligned, solder the remaining 20 leads. Take the time to keep every tube at the same height — the result reads much more cleanly when the row is uniform.

Install two 12-pin male headers on each tube board.

7.4 Step 3 — Final Assembly

Insert the RGB LEDs (LED1–LEDX) into the main board but do not solder them yet. Note orientation: the long lead(s) go toward the dot on the solder mask.

Plug each tube board into its corresponding female header on the main board.

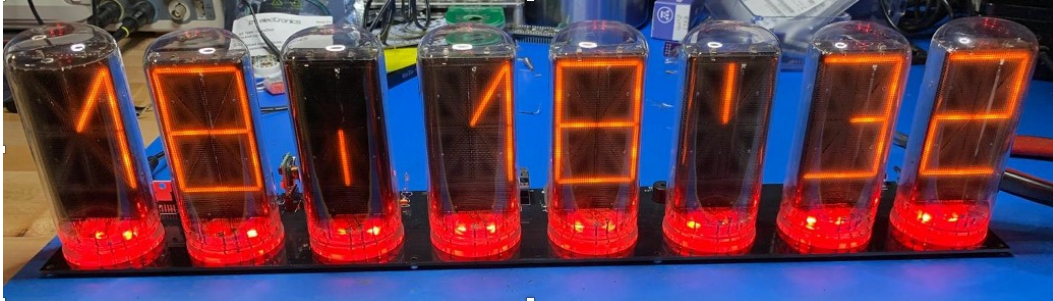
Turn the main board over and adjust each RGB LED so it sits about 1 mm below the bottom of its IV-17 tube, then solder all four leads.

Install the microcontroller, orienting the USB connector and antenna with the silkscreen marks. Insert the SD card.

Install the 10 mm standoffs in the bottom-edge holes of the main board with the supplied screws.

Power on and proceed to the Operator Manual for setup.

8. WordClock 4 — Nixie B7971



Finished WordClock 4 (8-digit, Burroughs B7971 Nixie tubes).

WordClock 4 uses Burroughs B7971 alphanumeric Nixie tubes and a high-voltage power chain producing approximately 170 V DC. The board uses three (6-tube) or four (8-tube) HV5530 high-voltage shift registers driven through a CD4504 level shifter. Up to two TaylorEdge 1363 boost converter modules supply the high-voltage rail.

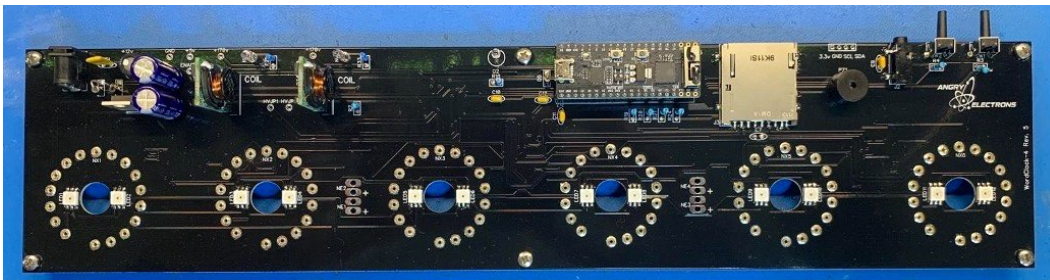
WARNING: The high-voltage circuit on this board is lethal. Read Section 4 (Safety Warnings) before applying power. Always disconnect the supply and wait 30 seconds for the storage capacitors to discharge before working on the board.

8.1 Parts List

Reference	Qty	Part	Value	Notes
C1, C2	2	Capacitor	330 μ F 25–50 V	Electrolytic
C3–C6, C8, C13	6 / 7	Capacitor	100 nF	SMD; C14 added on 8-tube
C7, C9–C12	5	Capacitor	100 nF	Through-hole
D1	1	Diode	1N4007	
D2, D5	2	Diode	1N5819	Schottky
F1	1	Poly Fuse	0.5–1 A	
J1	1	Power Jack	2.1 mm	
J2	1	GPS Jack	SJ-3523-SMT-TR	3.5 mm SMD
J3	1	SD Card Jack	DM1AA-SF-PEJ	
L1	1	Inductor	100 μ H	
LED1–LED12	12 / 16	RGB LED	APA102	Per-tube backlight
PZ1	1	Buzzer	—	
Q2	1	Sensor	TEPT4400	Light sensor
R1	1	Resistor	7.5k	8-tube only
R3, R4, R7, R19, R20	5	Resistor	10k	
R2, R33	2	Resistor	270k	
R5, R8	2	Resistor	330 Ω	8-tube only

R6, R34	2	Resistor	47k	
R18	1	Resistor	120k	
R21, R24	2	Resistor	4.7k	
R100 series	12 / 16	Resistor SMD	22k (1210/1206)	
R200 series	24 / 32	Resistor SMD	24k (1210/1206)	
R300 series	42 / 56	Resistor SMD	27k (1210/1206)	
R400 series	12 / 16	Resistor SMD	33k (1210/1206)	
R500 series	4	Resistor SMD	270k (1210/1206)	6-tube only
SW1, SW2	2	Pushbutton	90°	
U1	1	IC	CD4504	Level shifter
U2	1	IC	LM2576T-5.0	5 V regulator
U3	1	IC	74HC14D	8-tube only
U4	1	IC	74AHC125D	6-tube only
U5	1	Microcontroller	ESP32 PICO-KIT	Pre-programmed
HV1–HV3	3	IC	HV5530PGG-G	High-voltage shift register
HV4	1	IC	HV5530PGG-G	8-tube only
HVPS1, HVPS2	2	PSU module	TaylorEdge 1363	One sufficient for 6-tube; two required for 8-tube
NE1–NE4	4	Neon bulb	—	6-tube only
NE7, NE8	1 / 2	Neon bulb	—	1 for 6-tube, 2 for 8-tube
—	102 / 136	Socket pin	B7971	
—	6 / 8	Standoff	10 mm	
—	6 / 8	Screw	M3 8 mm	
—	6 / 8	Acrylic diffuser	—	
—	2	Brass tube	1.5 × 90 mm	6-tube only
—	2	Brass tube	1.5 × 50 mm	6-tube only
—	1	Header strip	1×40 female	Cut for U5
—	1 / 2	Header	2×6 female	PSU connectors
—	1	SD card	—	Pre-loaded

8.2 Step 1 — Build the Main Board



WordClock 4 main board (top), showing the Nixie tube socket positions and TaylorEdge 1363 PSU mounting pads.

Install the female pin sockets on top of the board. Use masking tape to hold them in place while soldering on the bottom.

Cut the single-row 40-pin female header into two 17-pin pieces and install at the U5 location.

Install two 2×6 female headers at the HVPS1 and HVPS2 locations.

Do NOT solder the microcontroller to the board.

Build the 5 V supply. Install J1, F1, C1, C2, C3, D1, D2, L1, and U1. C1, C2, D1, and D2 must be oriented per the silkscreen.

Connect a 12–18 V supply to J1 and check the test point for 5 V. Do not continue if 5 V is not present.

Build the high-voltage supply. Install R1, R2, R6, R7, R33, R34, NE7, and NE8.

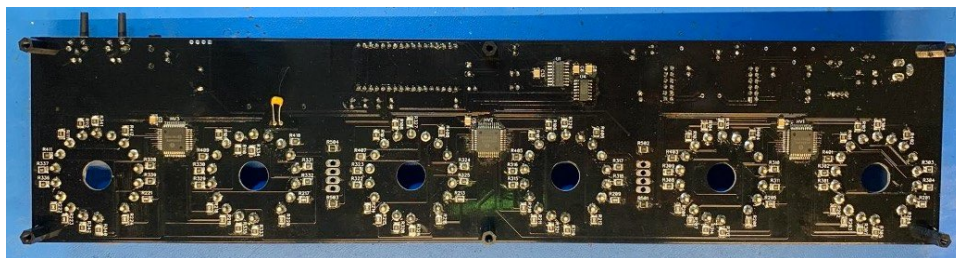
WARNING: Connect a temporary jumper between ENABLE and +5 V to enable the high-voltage supplies. Do NOT install the microcontroller while the jumper is in place or while testing the supplies — the ESP32 will be damaged.

Install one or two TaylorEdge 1363 high-voltage power supply modules with the coils facing the word "COIL" on the board. If only one supply is used, also connect a jumper between HVJP1 and HVJP2.

CAUTION: There is no load-balancing facility. Do NOT connect the HVJP1–HVJP2 jumper if two power supplies are installed — the supplies will fight each other.

Connect the 12 V supply. Both neon bulbs should light. Check one or both +170 V test points and confirm approximately +170 V DC.

If voltages are correct and the neons are lit, remove the power supplies and the ENABLE jumper.



WordClock 4 main board (bottom), showing the HV5530 high-voltage shift registers and supporting SMD parts.

Install the components on the bottom of the board, starting with the SMD resistors. Each "00 series" group is the same value — install all the R100 series first, then the R200 series, etc., to minimize the chance of a misplaced part.

Install the SMD capacitors next, then the ICs. The line or dot on each chip aligns with the cutout on the silkscreen footprint.

The HV5530s are notoriously difficult to solder cleanly. Inspect every joint under magnification — every pin must be soldered, with no bridges. Use solder wick to remove any bridges. A single bridged pin can take down a column of digits.

Install the SD card connector. All leads must be soldered to the board. The pins close to the case do not need to be soldered to the case. Install J3 (GPS connector).

Install the remaining top-side parts. The APA102 RGB LEDs install with the cutout aligned to the dot on the silkscreen.

Install Q2 (light sensor) about 10 mm above the board, shorter lead toward the back.

8.3 Step 2 — Final Assembly

Carefully install the Nixie tubes into the sockets. Reinstall the power-supply modules.

WARNING: Verify the ENABLE jumper has been removed before applying power.

Install the microcontroller, orienting the USB connector and antenna with the silkscreen marks. Insert the SD card.

Install six (6-tube) or eight (8-tube) 10 mm standoffs in the bottom-edge holes with the supplied screws.

Power on and proceed to the Operator Manual for setup.

9. Bring-Up Checklist

Common to every variant. Walk through every item on the first power-up — it pays back the time many times over.

- Visually inspect every solder joint under magnification, especially on the SMD ICs.
- Look for solder bridges on the SOIC, PLCC, and 1206 / 1210 footprints.
- Verify every electrolytic capacitor polarity (C1, C2, C9, etc.).
- Verify every diode orientation (D1–D5).
- With power off, confirm no shorts between supply rails — VCC to GND, +10.5 V to GND (WC2), HV to GND (WC4).
- Confirm test-point voltages match the values stated for each variant's Step 1.
- On WordClock 4, verify both +170 V test points before installing tubes.
- Confirm the ESP32 boots and creates the WiFi access point WordClock-X-X (where X-X matches the model).
- Connect to the AP at 192.168.4.1, enter WiFi credentials, and save. The clock should reboot, sync NTP, and display the time.
- Press SW1 — the time should briefly become "Always On" for the configured Button Override duration.
- Hold SW2 for 5 seconds — the clock should restart in AP mode.

If any of the bring-up items fails, stop and consult the Troubleshooting chapter of the Operator Manual before continuing.

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Nixie variants of this clock contain a small amount of mercury and neon gas under low pressure and operate from an internal high-voltage supply (~170 V). VFD variants operate from a ~35 V anode supply. Operate the clock at the supplied voltage only; do not open or service the high-voltage circuitry.

WordClock — Designed by Mitchell Feig